

The Z-Screw is an innovative initiative, that helps facilitate an environmentally sustainable growth of plants in challenged environments. By providing structural support during the initial phases of plant growth, the Zeedle aims to mitigate key environmental challenges in targeted areas. Over time, the material will (gradually) biodegrade in harmony with the plant life cycle, leaving behind a robust network for the plant to contribute to long-term eco-system stability.

Features/benefits

- Made of decomposable biomaterial
- **Higher survivability rates:** Protection of seedling during sprout phase reduce number of seeds, resulting in more efficient resource use.
- **Low maintenance:** The biodegradable structure (gradually) degrade, eliminating the need for ongoing plant maintenance.
- Versatile: Adaptable to different contexts and types of environments.
- Low cost: The product is designed to be affordable, allowing broader deployment.

The Zeedle is designed to facilitate restoration of eelgrass along the Danish coastline. Eelgrass plays an important role in marine ecosystems, as it stores carbon and nutrients in the seabed, improves water quality, and provides habitat for fish and other marine life. Over the past century, the eelgrass coverage along the Danish coastline has significantly decreased, and natural recovery faces major challenges such as natural predation, human disturbances and strong water currents.

The Zeedle helps address these challenges by enabling eelgrass to be planted and protected from disturbances during its early growth stages. Once the plant reach a stage of self-sustain, the structure will gradually degrade, allowing the plant to continue natural growth on their own. In this way, the Zeedle supports the restoration of marine ecosystems and the recovery of Danish coastal waters.

The Z-Screw

- The Z-Screw has in all phases of development had large support from environmental organisations and stakeholder throughout its lifetime.
- I. Winner of Innovation in Engineering (June 2025) at DTU
- Supported by NIRAS
- Supported by DTU Skylab

